Explainable **artificial intelligence** (XAI) for interpreting the contributing factors feed into the **wildfire** susceptibility prediction model

A Abdollahi, B Pradhan - Science of the Total Environment, 2023 - Elsevier

… of a **deep learning** (DL) model that is developed for **wildfire** … suggested model's output for **wildfire** susceptibility mapping. We … model's decision to map **wildfire** susceptibility, pinpoint high-…

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Comparisons between **wildfire** and forest harvesting and their implications in forest **management**

DJ McRae, LC Duchesne, B Freedman… - Environmental …, 2001 - cdnsciencepub.com

… **review** we summarize empirical evidence that illustrates disparities between forest harvesting and **wildfire**… Hence the objective of this **review** is to provide a comparative analysis of the …

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A **review** of the relationships between drought and **forest fire** in the United States

JS Littell, DL Peterson, KL Riley, Y Liu… - Global change …, 2016 - Wiley Online Library

… We **review** relationships between drought and fire regimes in United States forests, fire-related drought metrics and expected changes in fire risk, and implications for fire **management** …

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Effects of **wildfire** suppression chemicals on people and the environment—a **review**

KD Kalabokidis - Global Nest: The International Journal, 2000 - ccfruitandnuts.ucanr.edu

… Fire **management** officers must select fire control **techniques** … of retardants versus mechanical fire control **techniques**, ii) use of … desirable and efficient than a policy of crisis **management**. …

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Perceptions of **bushfire** risk mitigation and biodiversity conservation: a systematic **review** of fifteen years of research

EC Moskwa, I Ahonen, V Santala… - Environmental …, 2016 - cdnsciencepub.com

… Our objective for this study was to **review** recent social research that explores public and … to **bushfire management**. To do this we undertook a systematic **review** of **bushfire management** …

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Wildfire and climate change adaptation of western North American forests: a case for intentional **management**

PF Hessburg, SJ Prichard, RK Hagmann… - Ecological …, 2021 - Wiley Online Library

… **management** as the planned application of silvicultural and prescribed fire treatments and managed **wildfire** to … (2021), we **review** the strength of evidence surrounding the usefulness of …

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Deep neural networks for global **wildfire** susceptibility modelling

G Zhang, M Wang, K Liu - Ecological Indicators, 2021 - Elsevier

… **Wildfire** susceptibility is of great importance to the prevention and **management** of global … (MLPs), have been widely used in **wildfire** susceptibility. Recently, deep neural networks (DNNs…

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Forest fire detection system using wireless sensor networks and **machine learning**

U Dampage, L Bandaranayake, R Wanasinghe… - Scientific reports, 2022 - nature.com

… Furthermore, to acquire more accurate fire detection, a **machine learning** regression model is proposed. Because of the primary power supply provided by rechargeable batteries with a …

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Deep learning and transformer approaches for UAV-based **wildfire** detection and segmentation

R Ghali, MA Akhloufi, WS Mseddi - Sensors, 2022 - mdpi.com

… and optimized **Deep Learning** methods to detect **wildfire** at … proposed to identify and classify **wildfire** using aerial images. … were employed to segment **wildfire** regions and determine the …

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National inquiry on **bushfire** mitigation and **management**

S Ellis, P Kanowski, R Whelan - 2004 - ffm.vic.gov.au

… for **bushfire** mitigation and **management**; this report suggests the following indicative national **bushfire** … The Inquiry encourages the **Bushfire** Cooperative Research Centre to **review** this …

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Understanding evacuation preferences and **wildfire** mitigations among Northwest Montana residents

T Paveglio, T Prato, D Dalenberg… - … Journal of **Wildland Fire**, 2014 - connectsci.au

… This paper presents **survey** research from Flathead County, MT, USA, that aims to help … Both these groups have specific knowledge or skills related to **wildfire management** or damage, …

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A **review** of the main driving factors of **forest fire** ignition over Europe

A Ganteaume, A Camia, M Jappiot… - … **management**, 2013 - Springer

… This study analyses the factors driving **forest fire** ignition in the Mediterranean region including the most common human and environmental factors used for modelling in the European …

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Situational influences of acceptable **wildland fire management** actions

K Kneeshaw, JJ Vaske, AD Bright… - Society and Natural …, 2004 - Taylor & Francis

… Data for this study were obtained from a mail **survey** sent to individuals over the age of 18 … For the three forests combined, 3131 individuals completed a one-page, on-site **survey** and …

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Wildfire management in the United States: the evolution of a policy failure

G Busenberg - **Review** of policy research, 2004 - Wiley Online Library

… **wildfire** damages could have readily been used to justify this alternative **wildfire management** … Furthermore, the **wildfire** suppression policy cannot be explained as a rational choice to …

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Forest fire occurrence prediction in China based on **machine learning** methods

Y Pang, Y Li, Z Feng, Z Feng, Z Zhao, S Chen… - Remote Sensing, 2022 - mdpi.com

… of **machine learning** algorithms for the **forest fire** prediction model. Other applicable **machine learning** … In addition, the ability of these **machine learning** algorithms to analyse spatial …

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A **machine learning**-based approach for **wildfire** susceptibility mapping. The case study of the Liguria region in Italy

M Tonini, M D'Andrea, G Biondi, S Degli Esposti… - Geosciences, 2020 - mdpi.com

… Therefore, it is important to separately assess **wildfire** … based on a **machine learning** (ML) algorithm to elaborate **wildfire** … this kind of approach for **wildfire** susceptibility and risk mapping. …

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Learning to coexist with **wildfire**

MA Moritz, E Batllori, RA Bradstock, AM Gill, J Handmer… - Nature, 2014 - nature.com

… a more sustainable coexistence with **wildfire**. We have learned a great … In this **Review**, we summarize research on fire-prone … and **management** in all aspects of the coupled **wildfire** SES …

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Landscape–**wildfire** interactions in southern Europe: implications for landscape **management**

F Moreira, O Viedma, M Arianoutsou, T Curt… - … **management**, 2011 - Elsevier

… In this paper we **review** the available scientific knowledge on … for defining landscape **management** guidelines and policies that … policy and landscape **management** responses that …

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Getting ahead of the **wildfire** problem: Quantifying and mapping **management** challenges and opportunities

CD O'Connor, MP Thompson, F Rodríguez y Silva - Geosciences, 2016 - mdpi.com

… to support operational **wildfire management** decisions, with a … safety and effectiveness of **wildfire management** operations. We use … In this **review**, we provide an overview of some of the …

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Wildfire detection using transfer learning on augmented datasets

MJ Sousa, A Moutinho, M Almeida - Expert Systems with Applications, 2020 - Elsevier

… **Wildfire** detection is a time-critical application as the difficulty … and the breakthroughs in **deep learning** in complex visual … coupled with data augmentation **techniques** tested under a …

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